

Hellenic Complex Systems Laboratory

Accompaniment Instruments Frequencies, Networks and Configurations

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Abstract

We introduce *AccompaNet*, a Wolfram Language application linking instrument frequencies, co-occurrence networks and complete accompaniment configurations. Applied to a listening-based corpus of 100 recordings of 21 Smyrnic songs, it distinguishes instrument recurrence from partnership diversity. Guitar is the most prevalent and broadly connected resource, while cimbalom and tambourine exhibit contrasting patterns of recurrence and connectivity. Groups of instruments linked pairwise need not occur together, demonstrating the importance of retaining complete configurations. *AccompaNet* provides a reproducible framework for describing accompaniment composition and identifying comparisons for closer study of rhythmic articulation, instrumental texture and musical roles.

Keywords: *AccompaNet*; Smyrnic song; rhythm accompaniment; musical instruments; co-occurrence networks; accompaniment configurations

1. Introduction

Accompaniment configurations document the resources available for rhythmic articulation and the organization of instrumental texture in recorded performances. Their recurrence and variation are musicological concerns, although category presence alone does not establish how an instrument is played. Frequency, partner diversity and complete configurations distinguish resources recurring in restricted settings from those distributed across varied combinations, and establish whether a network pattern represents an attested accompaniment.

We introduce *AccompaNet*, a Wolfram Language application that connects these representations through interactive measures and configuration queries. We apply it to a listening-based corpus of 100 recordings of 21 Smyrnaic songs. Parts of this work were previously presented in C. Chatzimichail's Bachelor's Thesis and in the authors' Wolfram Demonstration (1,2). The present study develops the software and combines category prevalence, network analysis and configuration evidence to examine accompaniment composition.

Network approaches provide a useful framework for studying instrumentation. Percino and colleagues used a bipartite network of musical styles and instruments to examine instrumentation across a large recorded-music database (3). Here, the units are accompaniment categories and their combinations in a selected listening corpus. This scale permits direct comparison between a category's network position and the configurations that generate it.

The Ottoman contexts of Smyrnaic song and its retrospective national and stylistic classifications require historically situated interpretation (4). In Greek rebetika, heterophonic melodic lines, drones, rhythmic bass patterns and chords provide distinct forms of accompaniment (5). Instrument combinations consequently require interpretation in relation to repertoire and playing practice before they can be assigned a musical function or stylistic identity.

Complementary representations have also been used to investigate jazz musicians and bands (6) and musical affiliations at multiple scales (7). In the present study, nodes represent accompaniment categories and links record their presence in the same coded accompaniment. The configuration inventory supplies the group membership underlying those links.

We examine which accompaniment resources recur most often, how widely they combine with other resources, and which complete combinations are attested in this corpus. Frequency, network and configuration analyses address these complementary questions; sensitivity analyses assess dependence on rare links and individual configuration types.

2. Materials and methods

2.1. Corpus

The corpus comprises 100 recordings of 21 Smyrnaic songs selected from the Greek Music Archive of the Department of Popular and Traditional Music at the Technological Educational Institute of Epirus. Song selection was assisted by the musician Kyriakos Gouventas. Accompaniment categories were identified through aural analysis, and recordings were grouped by their accompaniment configurations.

The thesis summary figures classify 97 recordings in nine-beat metres, two in 4/4 and one in 7/4. The recordings in 4/4 and 7/4 represent two songs, each of which also has four nine-beat recordings.

The source thesis documents rhythmic variants across recordings of the same songs (1). Accompaniment configuration is therefore treated as a property of the recorded version. *AccompaNet* analyzes the aggregate configuration frequencies, retaining category membership and instrument multiplicity alongside a binary presence representation. This secondary analysis undertakes no new listening or instrument identification.

2.2. Accompaniment categories and configurations

The dataset contains 60 accompaniment configurations and 30 categories, with frequencies summing to 100 recordings. Configurations preserve instrument multiplicity: the 18 recordings with one guitar and one with two guitars form distinct configurations with the same category-presence set. The inventory therefore contains 59 distinct presence sets.

Bass and double bass are separate categories, occurring in 10 and 2 recordings, respectively. Synthesizer with drum timbre and synthesizer with toumperleki timbre are also distinct, each occurring once.

The guitar–toumperleki–baglamas–tambourine configuration occurs once.

The configuration data use the same category names as the interface. The complete category list is provided in Appendix, Section 8.4.

A node represents a coded accompaniment category. Single-label accompaniment denotes one distinct category and does not imply a solo performance. Configuration membership refers to the coded accompaniment rather than every sound source in a recording.

2.3. Incidence representation and pairwise network

Let the 60 multiplicity-preserving configurations be indexed by $c = 1, \dots, C$, with $C = 60$, and let the accompaniment categories be indexed by $i = 1, \dots, K$, with $K = 30$. Configuration c has the category-presence set S_c , its frequency is f_c , and its number of distinct labels is $m_c = |S_c|$. Let $\mathbf{1}\{\cdot\}$ denote the indicator function, equal to 1, when a specified condition holds and 0 otherwise.

We define the binary incidence matrix by

$$B_{ci} = \mathbf{1}\{i \in S_c\}, N = \sum_{c=1}^C f_c = 100. \quad (1)$$

The prevalence n_i and single-label count q_i are

$$n_i = \sum_{c=1}^C f_c B_{ci}, q_i = \sum_{c=1}^C f_c B_{ci} \mathbf{1}\{m_c = 1\}. \quad (2)$$

The weighted projection connects distinct labels that share at least one configuration:

$$w_{ij} = \sum_{c=1}^C f_c B_{ci} B_{cj} (i \neq j), w_{ii} = 0. \quad (3)$$

The binary adjacency is $A_{ij} = \mathbf{1}\{w_{ij} > 0\}$. Edge weights count recordings containing both categories, including their occurrence in larger configurations. Graph density is the number of distinct edges divided by $\binom{K}{2}$.

The graph has no self-loops. Prevalence and single-label frequency are reported separately from partnership measures. Single-label configurations remain in the incidence data and contribute to prevalence, although they generate no pairwise links.

2.4. Descriptive measures and group-size normalization

Degree $k_i = \sum_{j \neq i} A_{ij}$ measures the number of distinct partners. Weighted strength $s_i = \sum_{j \neq i} w_{ij}$ measures accumulated partner occurrences. Their relationship to complete configurations is

$$s_i = \sum_{c=1}^C f_c B_{ci} (m_c - 1). \quad (4)$$

An appearance in a five-category configuration contributes one to prevalence and four to strength. To account for configuration size, we also calculate a fractional projection using event-size weighting from affiliation networks (8):

$$\tilde{w}_{ij} = \sum_{c: m_c > 1} \frac{f_c B_{ci} B_{cj}}{m_c - 1} (i \neq j), \tilde{w}_{ii} = 0. \quad (5)$$

Summing this weight over partners gives the exact identity

$$\tilde{s}_i = \sum_{j \neq i} \tilde{w}_{ij} = n_i - q_i. \quad (6)$$

Each appearance with at least one other category contributes one unit to fractional strength. Raw strength counts accumulated partner occurrences, whereas fractional strength counts participation in multi-label configurations.

Binary betweenness measures the fraction of shortest paths between other labels that pass through a node, normalized over all unordered pairs of other nodes:

$$b_i = \frac{2}{(K-1)(K-2)} \sum_{u < v; u, v \neq i} \frac{\sigma_{uv}(i)}{\sigma_{uv}}. \quad (7)$$

Here, σ_{uv} is the number of shortest paths from u to v , and $\sigma_{uv}(i)$ counts those traversing i ; disconnected pairs contribute zero. Path-based measures are calculated on the binary graph, assigning unit length to every link.

For each pair, Jaccard overlap is $J_{ij} = w_{ij} / (n_i + n_j - w_{ij})$. It is the fraction of recordings containing either category that contain both. This normalization complements the raw co-occurrence count when category prevalences differ. Preferential association would require comparison with a null model that accounts for participation heterogeneity (9).

2.5. Configurations and projection loss

The incidence matrix and frequency vector define a weighted bipartite representation linking configuration types to categories, equivalently a collection of weighted hyperedges (9). The configuration inventory additionally retains instrument multiplicity.

One-mode projection loses information about the events generating edges (10), and different higher-order structures can yield the same pairwise network (11,12). We quantify one consequence of this loss by comparing graph triangles with jointly observed triples. A graph triangle is an unordered triple with all three pairwise links present. An observed triple is a three-category subset of at least one configuration, including configurations with four or five categories. We enumerate both sets and count the graph triangles without a corresponding observed triple.

Configuration queries return the C01–C60 identifiers and frequencies of all configurations containing the requested categories and multiplicities. Support is the sum of those frequencies. Worked examples were selected after inspecting the results: cimbalom and tambourine illustrate prevalence and partnership diversity, while guitar–bendir–toumperleki and bendir–oud–toumperleki illustrate the distinction between a projected triangle and a jointly observed triple.

2.6. Sensitivity and computational verification

Three analyses assess sensitivity to network structure and configuration composition. First, we retain edges observed at least two, three or four times, keeping all 30 nodes. Second, we remove guitar from the graph and examine the remaining components. Third, we delete each of the 60 configuration types, including all its repetitions, and recompute the five principal measures. Nodes remain fixed across deletion scenarios. Rank is one plus the number of categories with a strictly larger value.

AccompaNet constructs the incidence representation and loop-free graph, provides 17 selectable measures, and retrieves configuration evidence. Its Wolfram Language report computes graph summaries, projection loss and sensitivity analyses. Exact rational betweenness calculations agree with the native graph function for the full graph and all 60 configuration-deletion scenarios. Native evaluation verified the data, graph weights, configuration queries and selectable measures; all 51 combinations of measures and views constructed successfully. The network figure uses layout seed 20260925. Computational definitions, complete results and reproduction instructions are provided in the Appendix.

3. Results

3.1. Configuration size and instrument prevalence

The 100 recordings contain 198 category appearances. Thirty-seven recordings have one accompaniment category, 35 have two, 22 have three, five have four, and one has five, giving a mean of 1.98 distinct categories per recording. The 28 recordings with at least three categories contribute 106 of the 141 pair occurrences generated by projection (Figure 1; Table 1).

Guitar is the most prevalent category, appearing in 56 recordings, followed by *toumperleki* in 22, spoons in 14, *bendir* in 13, *baglamas* in 12, and bass in 10. Guitar is the only accompaniment category in 19 recordings and appears with other categories in 37. *Cimbalom* occurs in eight recordings and is the only coded accompaniment category in five.

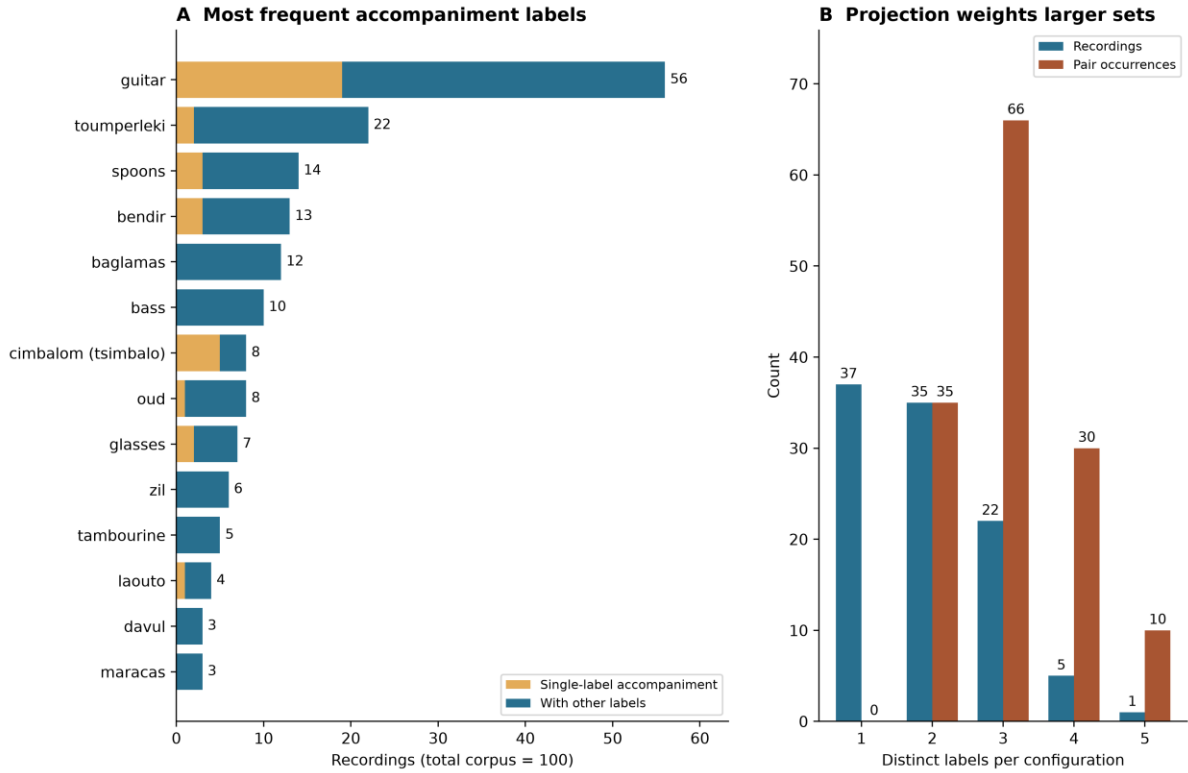


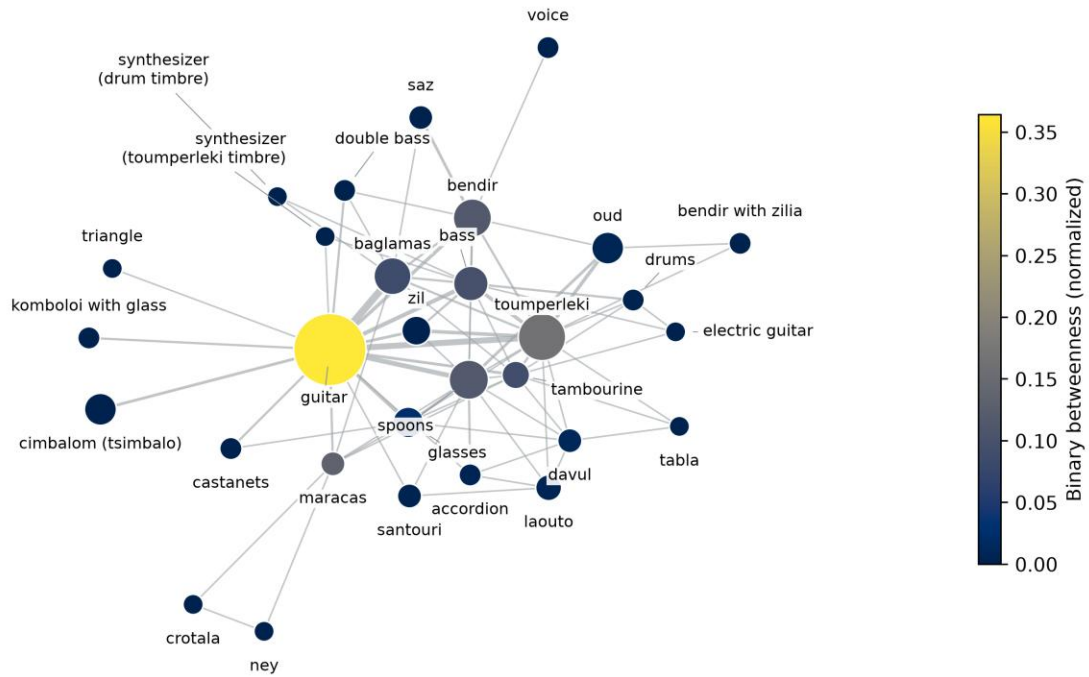
Figure 1. Prevalence and configuration size. (A) The 14 most frequent accompaniment categories, separated into single-label and multi-label appearances. (B) Recordings and pair occurrences contributed by each configuration size. Configurations with three or more categories account for 28 recordings and 106 of the 141 pair occurrences.

Table 1. Configuration sizes and their contributions to the pairwise projection. Size counts distinct accompaniment categories.

Labels per configuration	Counted recordings	Label appearances	Pair occurrences
1	37	37	0
2	35	70	35
3	22	66	66
4	5	20	30
5	1	5	10
Total	100	198	141

3.2. Pairwise structure and contrasting forms of centrality

The graph contains 30 nodes and 78 edges, with density 0.179. All categories belong to one connected component. The diameter is four links, and the mean shortest-path length is 2.147 (Figure 2).



Node area: recording prevalence; edge width: co-occurrence count
Positions are schematic; all 30 labels and 78 non-loop edges are shown.

Figure 2. Accompaniment co-occurrence network. The 30 nodes represent accompaniment categories, and the 78 edges represent observed pairs. Node area is scaled by prevalence with a minimum area for legibility; edge width increases with frequency. Node colour indicates normalized binary betweenness. Positions use a fixed-seed layout. Single-label accompaniment is reported separately from the loop-free graph.

Guitar has 17 distinct partners, followed by toumperleki with 14, spoons with 13, and bass and tambourine with 11 each. Guitar also has the largest weighted strength, 58, followed by toumperleki with 33 and bass with 22. Its normalized binary betweenness is 0.364, compared with 0.164 for toumperleki. Table 2 shows how prevalence, partner diversity and graph position diverge for other categories.

Maracas has the third-highest binary betweenness despite appearing in only three recordings. It occurs with ney and crotala in one configuration and connects to guitar in others, placing it on paths between this peripheral grouping and the rest of the network.

Raw and fractional strength also produce different emphases. Bass has higher raw strength than spoons, 22 versus 21, but lower fractional strength, 10 versus 11. Tambourine has raw strength 14 from only five appearances because it occurs with multiple partners. The two measures distinguish participation frequency from accumulated partner occurrences.

The highest-weight pair is guitar–baglamas, with ten co-occurrences. Guitar–toumperleki has eight, guitar–spoons six, and guitar–bass five. The corresponding Jaccard overlaps for guitar–baglamas

and guitar–toumperleki are 0.172 and 0.114. High instrument prevalence thus does not imply that a pair occupies a large fraction of the recordings in which either member appears (Figure 3).

3.3. Observed combinations and triangles created by projection

The 60 multiplicity-preserving configurations retain distinctions that the pairwise graph cannot show (Figure 4). They contain 59 distinct category-presence sets. The graph has 81 triangles, but only 51 distinct three-category subsets occur together within at least one configuration. The remaining 30 triangles, or 37.0%, arise from pairwise associations that are not jointly realized in any listed configuration.

Guitar, bendir and toumperleki form a graph triangle: guitar–bendir occurs in four recordings, guitar–toumperleki in eight, and bendir–toumperleki in two. No configuration contains all three categories. *AccompaNet* retrieves the separate configurations supporting each pair.

Bendir, oud and toumperleki, by contrast, occur together in configuration C53, with frequency one. This configuration directly supports their joint membership in a recorded accompaniment. Appendix, Section 8.8, provides the configuration evidence for both examples.

Table 2. Selected accompaniment categories and their descriptive network measures. Complete results for all 30 categories are provided in Appendix, Section 8.4. Fractional strength counts multi-label appearances. Binary betweenness is normalized as in Eq. (7).

Label	Prevalence	Single-label count	Degree	Strength	Fractional strength	Betweenness
Guitar	56	19	17	58	37	0.364
Toumperleki	22	2	14	33	20	0.164
Spoons	14	3	13	21	11	0.115
Bendir	13	3	9	17	10	0.116
Baglamas	12	0	9	21	12	0.086
Bass	10	0	11	22	10	0.097
Cimbalom (tsimbalo)	8	5	1	3	3	0.000
Oud	8	1	4	9	7	0.007
Tambourine	5	0	11	14	5	0.090
Maracas	3	0	7	8	3	0.135

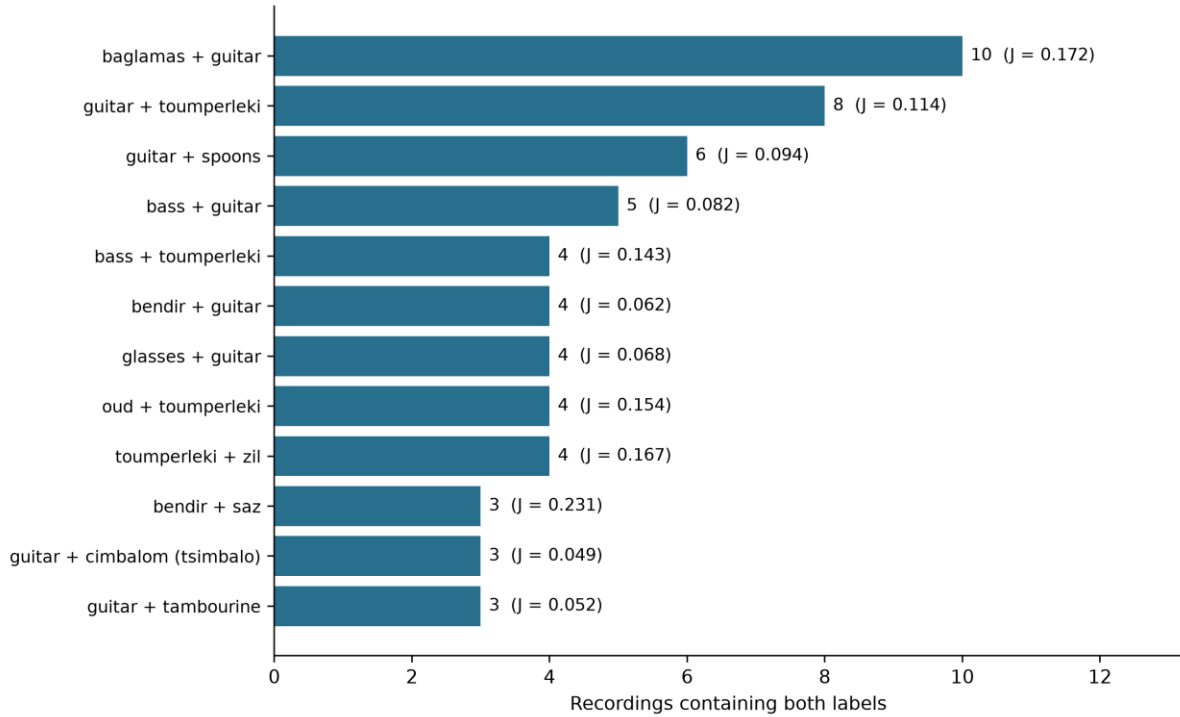


Figure 3. Co-occurrence counts and Jaccard overlap for the 12 most frequent pairs. Bar length gives the number of recordings containing both categories; annotations report the count and Jaccard overlap, J . Jaccard overlap normalizes co-occurrence by the number of recordings containing either category.

3.4. Dependence on rare links and individual configurations

Fifty-one of the 78 links, or 65.4%, have weight one. Restricting the graph to links observed at least twice leaves 27 edges, one connected component of 18 categories and 12 isolated categories. Higher thresholds reduce the connected core further (Table 3). Full connectivity is therefore sensitive to the inclusion of one-recording associations.

Table 3. Edge-frequency sensitivity with all 30 categories retained.

Minimum pair count	Retained edges	Isolated labels	Connected components	Largest component
1	78	0	1	30
2	27	12	13	18
3	14	18	19	12
4	9	21	22	9

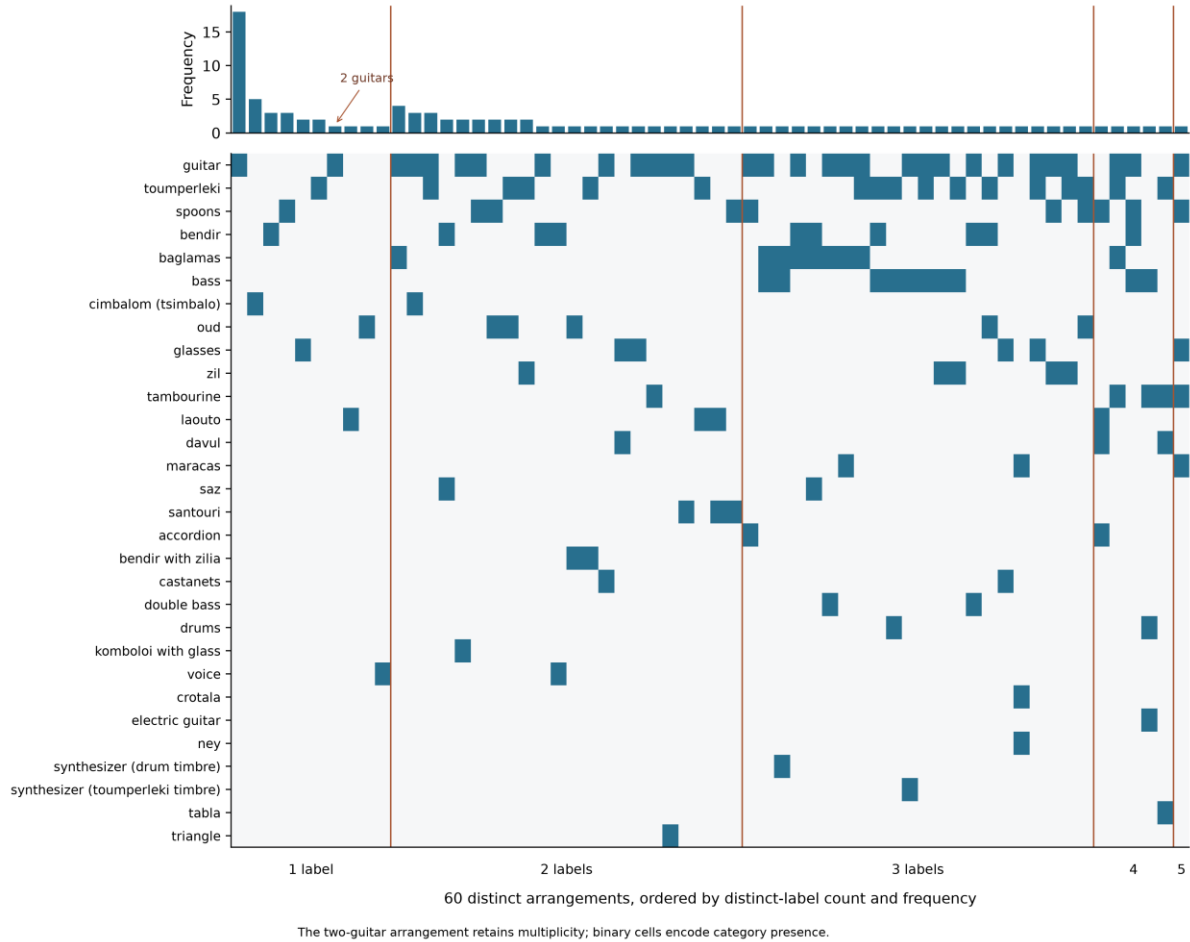


Figure 3. Binary incidence of 60 configurations across 30 accompaniment categories. Filled cells indicate membership, and upper bars show configuration frequencies. Columns are grouped by number of distinct categories, then by descending frequency. The marked two-guitar column has the same binary pattern as single guitar; their multiplicities and frequencies remain separate in the inventory. The 60 columns represent 59 distinct presence sets.

Removing guitar leaves 61 edges among 29 categories. Twenty-six categories remain connected, while cimbalo, komboloi with glass and triangle become isolated.

Across all 60 configuration deletions, guitar remains strictly first in prevalence, degree, raw strength, fractional strength and binary betweenness.

4. Discussion

4.1. What the network adds to the musicological description

Guitar's prominence identifies a recurrent resource across varied accompaniment settings. Its use as the only coded accompaniment category and with multiple partners does not specify its playing role. Pennanen documents guitar drones, rhythmic bass patterns and chordal support, and shows that chordal accompaniment can coexist with modal organization (5). Guitar's network position therefore provides no direct evidence of functional harmony or stylistic Westernization; these require musical and historical evidence.

Cimbalom and tambourine demonstrate why recurrence and partnership diversity should be considered separately. Cimbalom occurs in eight recordings, comprising five single-label accompaniments (C02) and three with guitar (C10), so its sole partner reflects a restricted set of combinations. Tambourine occurs once in each of five configurations (C22, C33, C43, C49 and C60), with eleven partners and fourteen accumulated partner occurrences. Four of these configurations contain at least four categories. Its broad connectivity reflects participation in diverse, relatively large combinations.

These distributions distinguish recurrent use within restricted accompaniment settings from infrequent participation in larger combinations. They do not measure audible prominence. Comparisons of cimbalom with and without guitar, and of tambourine across configurations, could examine how bass patterns, chordal support and pulse articulation are distributed among instruments.

The categories describe selected accompaniment resources rather than a uniform organological taxonomy. Named instruments, the composite bendir with zilia, voice and synthesizer timbres retain distinct positions in the source's listening-based classification. Their presence records available resources; it does not establish equivalent timbres or musical functions.

4.2. Why the complete configuration matters

The 30 graph triangles without a jointly observed triple show why network connectivity cannot establish an accompaniment type. Their pairs occur across different recordings, but the complete combination is unattested. Configuration evidence is therefore necessary before interpreting a connected group as a recorded instrumental combination; recurrence and musical analysis would be needed to establish a convention.

The guitar–bendir–toumperleki triangle is supported by separate pair occurrences, whereas C53 directly documents bendir, oud and toumperleki together. *AccompaNet* links network exploration to these underlying memberships. Retaining the inventory avoids discarding group information and subsequently attempting to reconstruct it (10–12).

Configuration size also affects pairwise weight. The 28 recordings with at least three categories generate 106 of 141 pair occurrences because each configuration contributes every constituent pair. Fractional weighting accounts for this effect at the category level by assigning one total unit per multi-label appearance. Its reduction to a participation count gives a direct interpretation alongside raw strength.

4.3. Stability and sparse associations

Guitar's leading position is stable across configuration deletions, while full graph connectivity depends on associations observed once. These results distinguish the robustness of guitar's ranking to configuration deletion from the dependence of overall connectivity on rare links.

An association observed once may still document a distinctive accompaniment choice. Frequency thresholds show how much network structure depends on such observations; their musical importance requires examination of the relevant recordings. The threshold analysis therefore complements the configuration inventory by identifying both recurring partnerships and combinations that merit individual attention.

Historical interpretation also requires attention to the repertoire's situated categories (4). Commercial selection, recording technology and reissue practices shape recorded evidence and can affect instrumentation and perceived sound (13). The pooled configurations consequently

document the selected recordings; they do not establish the prevalence of these combinations in live performance or a chronology of stylistic change.

4.4. Limitations and priorities for further work

The corpus is selected, and recordings of the same song may share musical material and performance conventions. The aggregate frequency table does not retain recording-to-song links, so the analysis cannot account for this dependence. Configuration deletion measures deterministic sensitivity and does not estimate sampling uncertainty.

The analysis uses aural accompaniment coding whose reliability has not been independently quantified. Recording-level presence does not identify accent patterns, bass movement, harmonic rhythm, simultaneous sounding or interaction. Shared nine-beat metre likewise does not imply identical rhythmic realization. These features require passage-level listening and transcription, beyond the present configuration inventory.

Extending the corpus to more recordings could support two complementary questions: why some instruments remain unobserved together and why particular combinations recur more frequently than others. Candidate explanations include timbral compatibility, practical convenience, performer availability and changing musical fashions. These are hypotheses for investigation. A missing pairwise link indicates that two categories never share a coded accompaniment in the selected corpus; an unattested complete combination may nevertheless contain pairs observed elsewhere. Absence within a specified corpus and historical period does not establish musical incompatibility or deliberate avoidance, particularly when instruments are rare or coverage is incomplete. Frequent co-occurrence alone does not demonstrate preference.

Further work should therefore link configurations to songs, performers, dates and archive identifiers, and assess annotation agreement. Comparisons of versions of the same song could then relate changes in instrumentation to bass patterns, chordal support, accentuation and tempo. Period comparisons would require the restored recording links; tests of preferential pairing would also require a justified null model (9). Such comparisons should account for instrument prevalence, configuration size and repeated versions of the same song. Listening, transcription and musicians' accounts could help assess whether the patterns are consistent with timbral, practical or stylistic explanations.

AccompaNet can support analyses of other recording collections by replacing its supplied data with user-defined lists of accompaniment configurations and their recording frequencies. Replacement lists should use consistent instrument labels and preserve any instrument multiplicities, and the relevant calculations should be re-evaluated. The core incidence, network and configuration analyses can thus be applied to new material; corpus-specific checks and sensitivity analyses should be adapted to the replacement dataset.

5. Conclusion

AccompaNet provides a differentiated account of accompaniment resources: which recur, how broadly they combine, and which complete combinations are attested. In this Smyrnaic corpus, guitar leads across prevalence and network measures, while cimbalom and tambourine distinguish recurrence from partnership diversity. Thirty of 81 network triangles lack a jointly observed triple, making configuration evidence essential to the interpretation of instrumental combinations. The resulting comparisons support focused study of accompaniment roles, while leaving their rhythmic, harmonic and historical interpretation to analysis of the recordings.

6. Acknowledgements

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8. Appendix

AccompaNet examines category prevalence, pairwise networks and complete accompaniment configurations. The study builds on C. Chatzimichail's thesis and the authors' Wolfram Demonstration (1,2).

8.1. Dataset

The dataset represents 100 recordings of 21 songs, grouped into 60 configurations across 30 categories. Each configuration retains repeated instruments: single guitar and two guitars are different arrangements but have the same category-presence set. The inventory therefore contains 59 distinct presence sets.

The 19 guitar-only recordings comprise 18 single-guitar recordings (C01) and one two-guitar recording (C24). The guitar–baglamas–toumperleki–tambourine quartet (C60) occurs once. C. Chatzimichail confirmed these frequencies for the publication. Bass and double bass remain separate, as do the two synthesizer timbres.

8.2. Computational definitions

Equations (1)–(7) in the main manuscript define the principal measures. Configuration size counts distinct categories. Repeated guitar affects arrangement identity but adds no binary incidence or self-loop. The incidence matrix has 60 rows and 30 columns; its loop-free projection has 78 edges. Frequency weighting gives 198 category appearances and 199 category entries when the second guitar is counted. The implementation satisfies

$$\sum_i n_i = \sum_c f_c m_c = 198, \sum_{i < j} w_{ij} = \sum_c f_c \binom{m_c}{2} = 141.$$
$$\sum_i s_i = 282, \tilde{s}_i = n_i - q_i.$$

For the binary, loop-free graph, harmonic centrality uses shortest-path distance $d(i, j)$ and is given by

$$h_i = \frac{1}{K-1} \sum_{j \neq i} \frac{1}{d(i, j)},$$

where an unreachable category contributes zero and the fixed denominator is $K - 1 = 29$. Normalized betweenness divides the unordered-pair total by 406, the number of pairs among the other 29 categories. Transitivity is three times the triangle count divided by the number of connected triples. Mean local clustering averages the fraction of linked neighbour pairs across all categories, assigning zero to those with fewer than two neighbours.

8.3. Projection and sensitivity

Triangle support is checked against every three-category subset of every configuration, including larger configurations. Of 81 network triangles, 51 are supported and 30 are unsupported (37.04%): their three categories never occur together in any configuration.

Sensitivity analyses retain the full category set while increasing the minimum pair count from 1 to 4 or deleting each complete configuration and all its repetitions. C01 and C24 are deleted separately. Guitar remains strictly first in prevalence, degree, raw strength, fractional strength and normalized betweenness in all 60 configuration-deletion scenarios. A separate analysis removes the guitar

vertex and its incident edges. These are deterministic sensitivity checks, not estimates of sampling uncertainty.

8.4. Category results

The table uses the category names stored directly in the notebook. Bass and double bass are separate categories, as are the two synthesizer timbres. Single-label counts include both C01 and C24 for guitar; betweenness uses the normalization in Section 8.1.

Category	Prevalence	Single-label	Degree	Strength		Fractional strength	Betweenness
guitar	56	19	17	58		37	0.364082
toumperleki	22	2	14	33		20	0.164316
spoons	14	3	13	21		11	0.114848
bendir	13	3	9	17		10	0.115742
baglamas	12	0	9	21		12	0.085649
bass	10	0	11	22		10	0.096864
cimbalom (tsimbalo)	8	5	1	3		3	0.000000
oud	8	1	4	9		7	0.007471
glasses	7	2	7	10		5	0.024133
zil	6	0	4	10		6	0.000000
tambourine	5	0	11	14		5	0.090443
laouto	4	1	5	5		3	0.005927
davul	3	0	7	7		3	0.012728
maracas	3	0	7	8		3	0.134691
saz	3	0	2	4		3	0.000000
santouri	3	0	3	3		3	0.003577
accordion	2	0	4	5		2	0.006129
bendir with zilia	2	0	2	2		2	0.000000
castanets	2	0	2	3		2	0.000000
double bass	2	0	3	4		2	0.000000
drums	2	0	4	5		2	0.002463
komboloi with glass	2	0	1	2		2	0.000000
voice	2	1	1	1		1	0.000000
crotala	1	0	2	2		1	0.000000
electric guitar	1	0	3	3		1	0.000000
ney	1	0	2	2		1	0.000000

Category	Prevalence	Single-label	Degree	Strength		Fractional strength	Betweenness
synthesizer (drum timbre)	1	0	2	2		1	0.000000
synthesizer (toumperleki timbre)	1	0	2	2		1	0.000000
tabla	1	0	3	3		1	0.000000
triangle	1	0	1	1		1	0.000000

8.5. Configuration frequencies

Configurations C01–C60 retain instrument multiplicity and use the category names stored in the notebook. C24 contains two guitar entries. Frequencies sum to 100.

ID	Accompaniment labels	Frequency
C01	guitar	18
C02	cimbalom (tsimbalo)	5
C03	baglamas; guitar	4
C04	guitar; toumperleki	3
C05	spoons	3
C06	bendir	3
C07	guitar; komboloi with glass	2
C08	guitar; spoons	2
C09	guitar; santouri	1
C10	guitar; cimbalom (tsimbalo)	3
C11	oud; spoons	2
C12	oud; toumperleki	2
C13	glasses	2
C14	bendir; saz	2
C15	toumperleki	2
C16	toumperleki; zil	2
C17	accordion; davul; laouto; spoons	1
C18	accordion; guitar; spoons	1
C19	castanets; guitar	1
C20	guitar; triangle	1
C21	bendir; guitar	1
C22	guitar; tambourine	1

ID	Accompaniment labels	Frequency
C23	glasses; guitar	1
C24	guitar; guitar	1
C25	guitar; spoons; zil	1
C26	baglamas; guitar; maracas	1
C27	baglamas; bass; guitar	1
C28	baglamas; bendir; guitar	1
C29	baglamas; guitar; toumperleki	1
C30	bass; guitar; zil	1
C31	bass; bendir; guitar; spoons	1
C32	bendir; double bass; guitar	1
C33	glasses; guitar; maracas; spoons; tambourine	1
C34	guitar; toumperleki; zil	1
C35	bass; guitar; toumperleki	1
C36	glasses; guitar; toumperleki	1
C37	castanets; glasses; guitar	1
C38	laouto	1
C39	laouto; santouri	1
C40	laouto; toumperleki	1
C41	baglamas; double bass; guitar	1
C42	baglamas; bass; synthesizer (drum timbre)	1
C43	bass; drums; electric guitar; tambourine	1
C44	bass; bendir; toumperleki	1
C45	bass; toumperleki; zil	1
C46	bass; drums; toumperleki	1
C47	crotala; maracas; ney	1
C48	davul; glasses	1
C49	davul; tabla; tambourine; toumperleki	1
C50	oud	1
C51	bendir with zilia; oud	1
C52	oud; spoons; toumperleki	1
C53	bendir; oud; toumperleki	1

ID	Accompaniment labels	Frequency
C54	baglamas; bendir; saz	1
C55	spoons; santouri	1
C56	bass; guitar; synthesizer (toumperleki timbre)	1
C57	bendir with zilia; toumperleki	1
C58	voice	1
C59	bendir; voice	1
C60	baglamas; guitar; tambourine; toumperleki	1

8.6. Reproducing the analysis

Use AccompaNet.nb, version 1.0.0, dated 28 September 2026. The notebook contains the configuration data and all required calculations. It was checked in Wolfram Language 15.0.1 for Microsoft Windows (64-bit). Evaluate its initialization cells in saved order. Run `accompaNet[]` to open the Recording frequency and Supplementary analysis tabs. To inspect the report directly, run

```
report = accompaNetReport[];
```

```
accompaNetReportView[report]
```

The report provides category, pairwise and configuration tables, triangle support, graph summaries and all sensitivity analyses. `report["AllChecksPassed"]` returns True when the embedded checks pass. Individual checks are available in `report["Checks"]`. The numerical analysis uses no random sampling; graph layouts are schematic.

8.7. Configuration queries

`accompaNetEvidence[items]` lists matching configurations and frequencies; `accompaNetSupport[items]` sums their frequencies. Queries use the category names in Sections 8.4 and 8.5 and respect multiplicity. For example, `accompaNetSupport[{"guitar"}]` returns 56, `accompaNetSupport[{"guitar", "guitar"}]` returns 1, and three guitars return 0. Set the second argument to True for an exact configuration match: `accompaNetSupport[{"guitar"}, True]` returns 18. Unknown category names return \$Failed.

8.8. Worked examples

These examples were selected after inspecting the results. Configuration identifiers refer to Section 8.5.

8.8.1. Occurrence and partnership diversity

Cimbalom (tsimbalo) occurs in C02 five times and C10 three times: prevalence 8, single-label count 5 and one partner. Tambourine occurs once in each of C22, C33, C43, C49 and C60: prevalence 5, eleven partners and strength 14.

8.8.2. Pairwise support without joint occurrence

Guitar–bendir is supported by C21, C28, C31 and C32 (total 4); guitar–toumperleki by C04, C29, C34, C35, C36 and C60 (total 8); and bendir–toumperleki by C44 and C53 (total 2). No configuration contains all three categories. Pair counts include every larger configuration containing the pair.

8.8.3. Joint occurrence

C53 contains bendir, oud and toumperleki and occurs once. It records their joint presence, whereas the preceding triangle combines pairs from different configurations. Co-occurrence describes coded presence and does not establish acoustic synchrony or musical interaction.

8.9. Software availability and requirements

Software name: AccompaNet

Version: 1.0.0

DOI: [10.5281/zenodo.23063739](https://doi.org/10.5281/zenodo.23063739)

Project home page: <https://www.hcsl.com/Tools/MusicalNetworks/index.html>

Computational source notebook: *AccompaNet.nb*. Available at:
<https://www.hcsl.com/Tools/MusicalNetworks/AccompaNet.nb>

Programming language: Wolfram Language

Operating systems: Microsoft Windows 10+, Linux 4.18+, Apple macOS 13+

Other software requirements: [Wolfram Player](#) (freely available) or [Wolfram Mathematica](#).

Hardware requirements: x86-64-v2 and higher or Apple Silicon microarchitecture CPU, and 4 GB+ RAM, recommended.

Software citation: Chatzimichail C, Hatjimihail AT. *AccompaNet: Accompaniment Instruments Frequencies, Networks and Configurations*. Version 1.0.0. Hellenic Complex Systems Laboratory; 2026. DOI: <https://doi.org/10.5281/zenodo.23063739>

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AccompaNet.nb contains the configuration data, interactive measures and configuration queries used in this study. Its native report reproduces the numerical results and sensitivity analyses and includes computational checks. Appendix, Section 8.6, provides the Wolfram Language reproduction commands.

8.10. Citation

Chatzimichail C, Hatjimihail AT. *Accompaniment Instruments Frequencies, Networks and Configurations*. Technical Report XXX. Hellenic Complex Systems Laboratory; 2026. DOI: <https://doi.org/10.5281/zenodo.23067849>

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